

Linux Command Line Basics

30/11/2025

1. Introduction to the Linux Command Line

The Linux command line (often called the **terminal**, **shell**, or **CLI**) is a text-based interface that allows you to interact with your system quickly and efficiently. While graphical interfaces (GUIs) are great for daily tasks, the command line provides:

- Faster automation
- Power-user flexibility
- Access to advanced tools
- A universal interface across all distros

Most distributions ship with **bash** as the default shell, though others (zsh, fish) also exist.
To open a terminal:

- **Ubuntu / Debian:** Ctrl + Alt + T
 - **Fedora:** Activities → Terminal
 - **Any distro:** Search "Terminal"
-

2. System Information

Linux provides several commands to check system details:

2.1 `uname`

Shows basic system information:

```
$ uname  
$ uname -a  
$ uname -r  
$ uname -m
```

2.2 hostnamectl

Shows OS name, kernel, hardware:

```
$ hostnamectl
```

2.3 inxi

A more advanced tool (you may need to install it):

```
$ sudo apt update # update package index  
$ sudo apt install inxi  
$ inxi -F
```

Note: you need to update the package index before installing any package.

Displays CPU, GPU, RAM, kernel, drivers, etc.

2.4 lsb_release

Useful on Debian-based systems:

```
$ lsb_release -a
```

3. Shell Variables

The shell stores data (variables) temporarily during your session.

3.1 Viewing Variables

```
$ echo $HOME  
$ echo $USER  
$ echo $PATH
```

3.2 Creating Variables

```
$ NAME="Khaled"  
$ echo $NAME
```

3.3 Exporting Variables

Makes them available to programs:

```
$ export EDITOR=nano
```

3.4 Listing All Variables

```
$ printenv  
$ set
```

4. File System Navigation

Every Linux filesystem starts from the **root** (/) directory.

4.1 pwd – Print Working Directory

Shows where you currently are:

```
$ pwd
```

4.2 `ls` – List Files

```
$ ls
$ ls -l
$ ls -la
$ ls -h
```

4.3 `cd` – Change Directory

```
$ cd /home
$ cd Documents
$ cd ..
$ cd ~
$ cd -
```

Shortcuts:

- `.` → current directory
- `..` → parent directory
- `~` → home directory
- `-` → previous directory

5. Essential File & Directory Commands

5.1 `mkdir` – Create Directory

```
$ mkdir projects
$ mkdir -p school/assignments
```

5.2 `touch` – Create Empty Files

```
$ touch notes.txt
```

5.3 cp – Copy

```
$ cp file.txt backup.txt  
$ cp -r folder/ folder_backup/
```

5.4 mv – Move/Rename

```
$ mv oldname.txt newname.txt  
$ mv file.txt /home/user/Documents/
```

5.5 rm – Remove

Dangerous but essential

```
$ rm file.txt  
$ rm -r folder/  
$ rm -rf folder/ # very dangerous
```

5.6 cat – View Content

```
$ cat file.txt
```

5.7 echo – Print Text

```
$ echo "Hello Linux!"  
$ echo $USER
```

5.8 sleep – Pause Execution

```
$ sleep 5
```

6. Understanding Paths

6.1 Absolute Paths

Start from the root /.

Example:

```
/home/khaled/projects/code.py
```

6.2 Relative Paths

Start from your current location.

Example:

```
../images/photo.png  
./run.sh
```

7. Package Management

Every Linux distribution uses a package manager to install software.

7.1 apt – Debian, Ubuntu, Linux Mint

Update repositories:

```
$ sudo apt update
```

Install:

```
$ sudo apt install gedit
```

Remove:

```
$ sudo apt remove gedit
```

Upgrade system:

```
$ sudo apt upgrade
```

7.2 dnf – Fedora, RHEL 8+, CentOS Stream

```
$ sudo dnf install nano  
$ sudo dnf remove nano  
$ sudo dnf update
```

7.3 yum – Older RHEL/CentOS

```
$ sudo yum install nano  
$ sudo yum remove nano  
$ sudo yum update
```

8. Editors

8.1 nano (Terminal Editor)

```
$ nano file.txt
```

Controls:

- Save: **Ctrl + S**
- Exit: **Ctrl + X**
- Search: **Ctrl + W**

8.2 gedit (GUI Editor)

```
$ gedit notes.txt &
```

The & lets the terminal continue running.

9. Getting Help

Linux has multiple built-in help resources.

9.1 `man` – Manual Pages

```
$ man ls  
$ man tar
```

9.2 `info`

```
$ info coreutils
```

9.3 `tldr` – Simplified Examples

Install:

```
$ sudo apt install tldr
```

Use:

```
$ tldr ls
```

9.4 `--help`

```
$ ls --help  
$ tar --help
```

9.5 `apropos` – Search for Commands by Description

```
$ apropos network
```

10. Archiving & Compression

10.1 tar (Most Common in Linux)

Create:

```
$ tar -cvf archive.tar folder/
```

Extract:

```
$ tar -xvf archive.tar
```

Compressed with gzip:

```
$ tar -czvf archive.tar.gz folder/  
$ tar -xzvf archive.tar.gz
```

10.2 zip

```
$ zip archive.zip file1 file2  
$ unzip archive.zip
```

10.3 7z (7-Zip)

Install:

```
$ sudo apt install p7zip-full
```

Compress:

```
$ 7z a archive.7z folder/
```

Extract:

```
$ 7z x archive.7z
```

11. Bonus Useful Commands

11.1 `clear` – clears the terminal

11.2 `history` – shows previous commands

11.3 `head` / `tail` – preview files

```
$ head file.txt
$ tail file.txt
$ tail -f log.txt
```

11.4 `less` – view files with navigation

```
$ less file.txt
```

11.5 `grep` – search text

```
$ grep "error" log.txt
```

11.6 `whoami` – current user

11.7 `df -h` – disk usage

11.8 `du -sh folder/` – folder size

12. Final Tips for Newcomers

- Practice every day
- Use `tldr` for quick examples
- Avoid `rm -rf` unless absolutely necessary
- Use tab completion
- Use `history` to track commands
- Explore `/etc`, `/home`, `/usr`, `/bin` directories carefully

Practice Exercises

1. Create a directory structure:

```
projects/python/basics
```

2. Create three files inside `basics`.
3. Write your name into `info.txt` using `echo`.
4. Compress the folder using:
 - `tar`
 - `zip`
5. Install `tlldr` and test 5 commands.
6. View your system kernel version and CPU info using two methods.